

Data Tables

Signature of Lab Faculty:

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Date:

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**** For all the data tables, take data up to three decimal places, round to two, then enter into the table.**

Table 1: Resistance and Capacitance Data

For all your future calculations, please use the observed values only (even for theoretical calculations).

Notation	Expected Resistance	Observed Resistance (k Ω)	Notation	Expected Capacitance	Observed Capacitance (μF)
R	1 k Ω	0.987	C ₁	1 μF	1.09
			C ₂	1 μF	1
			C ₃	0.47 μF	0.436

Table 2: Data from Circuit 1 (Initial)

Keep the switch to the **initial position** (connect to 2V and keep 6V open).

Initial Circuit	Initial DC Supply Voltage $V_s(0)$ (V)			$v_c(0)$ (V)	$v_R(0)$ (V)	$I(0) = \frac{v_R(0)}{R}$ (mA)
	Expected Voltage	From DC power supply	Using multimeter			
Experimental	2.0	2.0	1.974	1.974	0	0
Theoretical				2	0	0

Table 3: Data from Circuit 1 (Final)

Change the switch to the final position (connect to 6V and keep 2V open).

Final Circuit	Final DC Supply Voltage $V_s(\infty)$ (V)			$v_c(\infty)$ (V)	$v_R(\infty)$ (V)	$I(\infty) = \frac{v_R(\infty)}{R}$ (mA)
	Expected Voltage	From DC power supply	Using multimeter			
Experimental	6.0	6.0	6.06	6.06	0	0
Theoretical				6	0	0

Table 4: Data from Circuit 2

Use function generator for the supply voltage and observe all values from the oscilloscope.

Time constant, $\tau = RC =$ 0.987 ms

Theoretical charging / discharging time = $5\tau = 5RC =$ 4.935 ms

Circuit 2	Supply Voltage V_s		Capacitor Voltage v_c		Charging / Discharging time t_{full} (ms)
	Minimum Value $V_{s_{min}}$ (V)	Maximum Value $V_{s_{max}}$ (V)	Minimum Value $v_{c_{min}}$ (V)	Maximum Value $v_{c_{max}}$ (V)	
Experimental (from oscilloscope)	1.92	6.40	2.08	6.32	4.86
Theoretical			2	6.5	5

Table 5: Data from Circuit 3

Use function generator for the supply voltage and observe all values from the oscilloscope.

Circuit 3	Charging Phase				Discharging Phase			
	Resistor Voltage		Circuit Current $I = \frac{v_R}{R}$		Resistor Voltage		Circuit Current $I = \frac{v_R}{R}$	
	$v_{R_{min}}$ (V)	$v_{R_{max}}$ (V)	I_{min} (V)	I_{max} (V)	$v_{R_{min}}$ (V)	$v_{R_{max}}$ (V)	I_{min} (V)	I_{max} (V)
Experimental	1.80	6.20	1.82	6.28	-3.80	4.20	-3.85	4.25
Theoretical	2	6.5	2	6.5	-4	4.5	-4	4.5

Table 6: Data from Circuit 4 & Circuit 5

Use function generator for the supply voltage and observe all values from the oscilloscope.

Peak to peak voltage to set on the function generator =

5 V

DC offset to set on the function generator =

0.5 V

Circuit 4	Capacitor Voltage v_C		Charging / Discharging time t_{full} (ms)	Time constant $\tau = \frac{t_{full}}{5}$ (ms)	Equivalent Capacitance		
	Minimum Value $v_{C_{min}}$ (V)	Maximum Value $v_{C_{max}}$ (V)			(from osc.) $C = \frac{\tau}{R}$ (μF)	From Circuit 5 using multimeter C_{eq} (μF)	Error $\frac{ C - C_{eq} }{C_{eq}} \times 100\%$ (%)
Experimental	-2.16	2.20	15.02	1.004	1.017	0.961	5.20%
Theoretical				1	1	1	

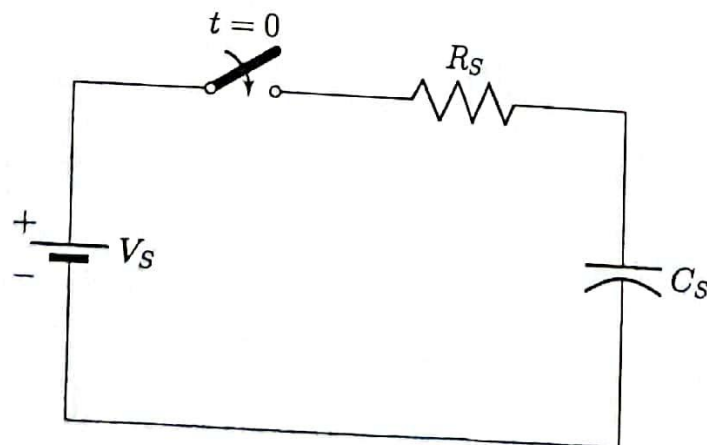
Questions

1. A capacitor stores energy-
☐ Magnetically ☒ Electrically ☐ Chemically ☐ Electro-chemically
2. If the capacitance (C) of a capacitor is related with the voltage (V) and the current (I) of the capacitor as $C = \frac{V}{I}$, which one of the following statements is correct? Capacitance of a capacitor can be increased by-
☒ increasing the applied voltage across the capacitor.
☐ increasing the charge accumulation on the plates of the capacitor.
☐ increasing the surface area of the plates.
☐ decreasing the size of the capacitor.
3. Under a certain dc bias, the time it takes for a particular capacitor to increase its voltage from 0 V to 5 V and from 2 V to 5 V in an RC circuit is-
☒ different ☐ same

Why?

Because charging 2V to 5V will be faster than 0V to 5V. It takes 5 times the time constant to charge 0V to 5V.

4. When the switch in the following circuit is closed at $t = 0$, the following energy conversions happen-



[use the keywords *electrical/mechanical/chemical/electro-chemical/thermal/heat* to answer (a) (b) and (c)]

- (a) The battery converts electro-chemical energy to electrical energy.
- (b) The capacitor receives electro-chemical energy from the battery and stores in the form of electrical energy.
- (c) The resistor dissipates energy into heat energy.
- (d) Upon being fully charged by the battery (not to be dead so quickly), the capacitor—
- ☐ spontaneously releases the stored energy after some time to the resistor connected.
 - ☐ gives the stored energy back to the battery after some time.
 - ☒ holds the energy until some other circuit elements are connected to receive it.
 - ☐ the capacitor can better tell what it wants to do.

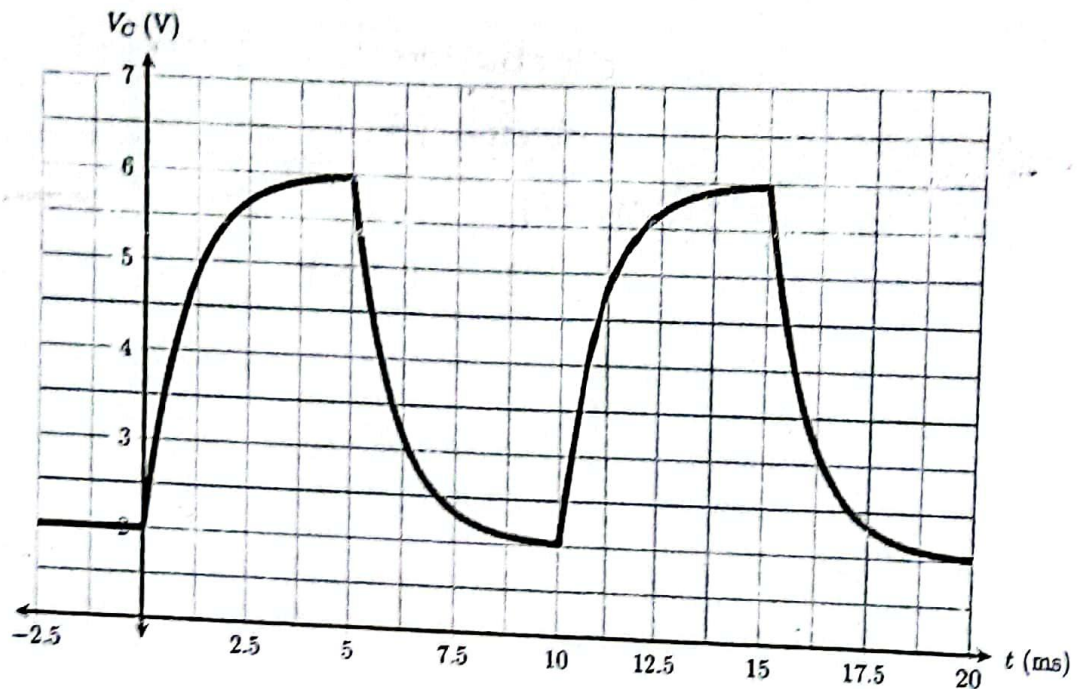
5. Why was it necessary to short the two terminals of a capacitor before measuring the capacitance in laboratory?

Because when a capacitor is charged, it stores electrical energy. The residual charge can affect the measurement and will give incorrect charge result.

6. Write briefly the significance of the time constant (τ) related to charging and discharging in a RC circuit?

The significance of τ is that understanding the transient behavior of an RC circuit during charging and discharging processes, it helps determine how quickly the capacitor approaches its final voltage during charging and how quickly it discharges.

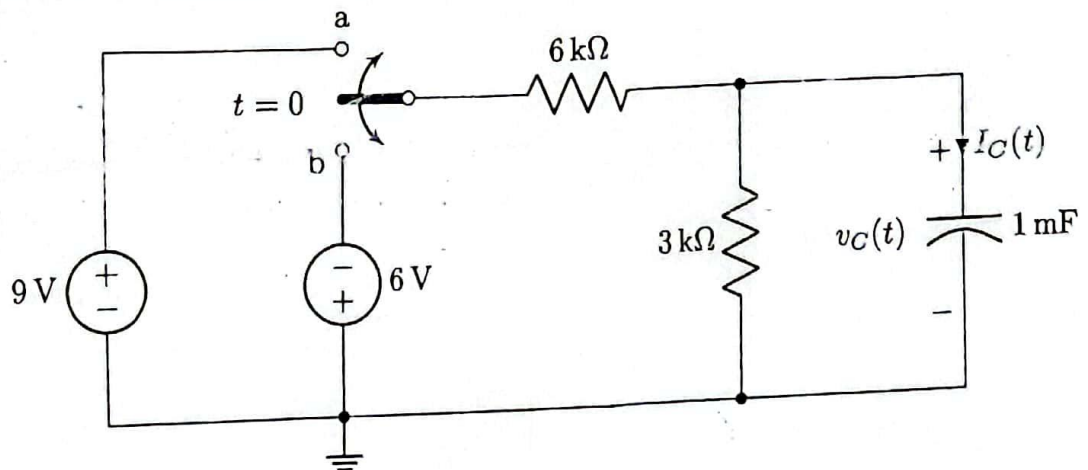
7. The capacitor voltage waveform you observed in the laboratory for Circuit 2 is given below.



- (a) Mark the following things in the diagram:
- (i) Charging and discharging phases (or timestamps)
 - (ii) The time constant (τ) for both charging and discharging phases
 - (iii) Initial and final voltages for both charging and discharging phases and
 - (iv) The times where the capacitor gets fully charged and discharged.
- (b) Explain how can you increase the duty cycle of the voltage waveform?

Adjusting the timing we can increase the duty cycle of the voltage waveform.

8. Consider the RC circuit shown below. At $t = 0$, the switch starts to alternate between positions a and b at a frequency of 500 Hz.



- (a) Which one of the following instruments do you need in laboratory for setting up the switching operation as shown in the circuit diagram above?

- ☐ Two separate DC Power supplies.
- ☐ A Function Generator with the functionality of providing a dc offset.
- ☐ An Oscilloscope.
- ☒ A DC Power Supply with two channels.

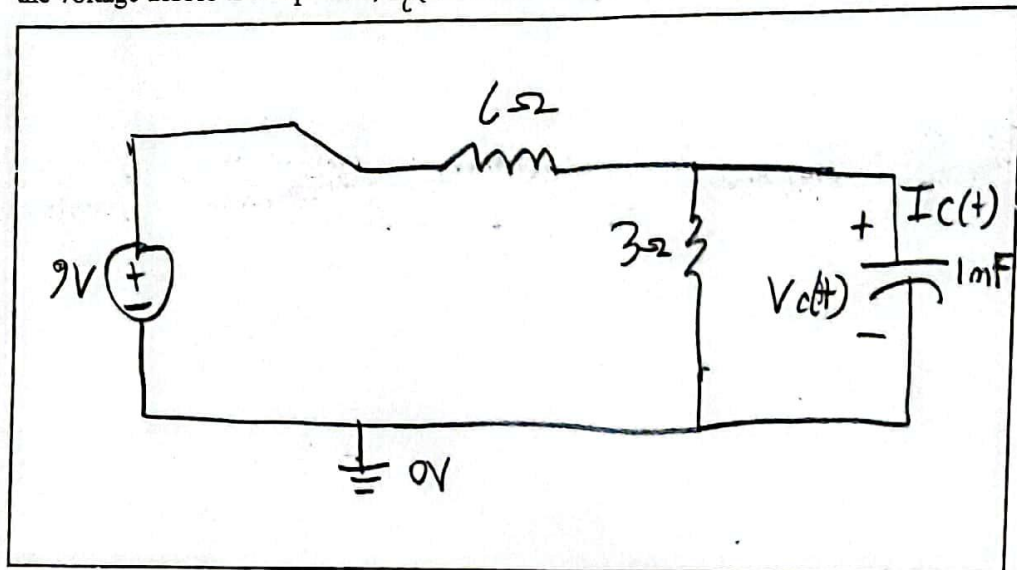
- (b) Based on your selection in (a), list down the values (amplitude of voltage/frequency/offset/scale etc.) that you will set on the selected instrument.

As I will use (instrument name:) DC Power supply with two channels to model the switching mechanism as described in the question statement, I will need to set only the (list:) Voltage on the instrument as follows (leave blank if not required to set any):

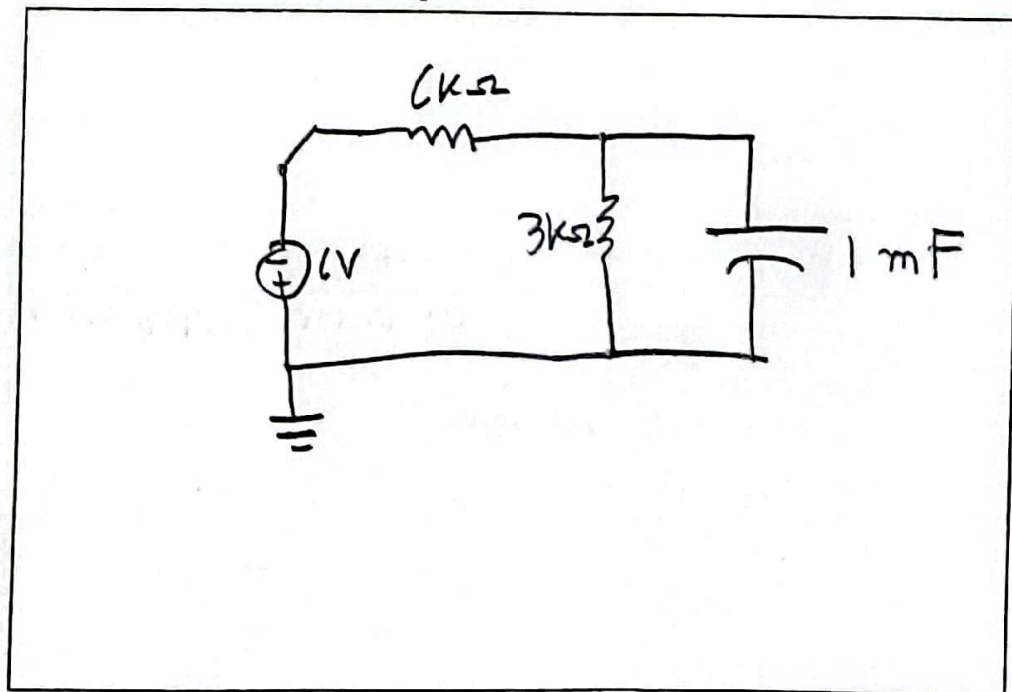
1. Amplitude of the voltage:
2. Frequency:
3. DC Offset:
4. Others (specify with values):

	V
	Hz
	V

- (c) Draw the active portion of the circuit when the switch is in position *a* and determine the voltage across the capacitor, $v_c(t, \text{switch} \rightarrow a)$.



- (d) Draw the active portion of the circuit when the switch is in position *b* and determine the voltage across the capacitor, $v_c(t, \text{switch} \rightarrow b)$.



- (e) So, the capacitor voltage $v_c(t)$ alternates between the values 9 (V) and 6 (V).

- (f) In general, the voltage across a capacitor under a sudden change in the applied dc bias is,

$$v_c(t) = v_c(\text{final}) + [v_c(\text{initial}) - v_c(\text{final})]e^{-\frac{t}{\tau}} \text{ or}$$

$$v_c(t) = v_c(\infty) + [v_c(0) - v_c(\infty)]e^{-\frac{t}{\tau}}$$

So, when $v_c(\text{final}) > v_c(\text{initial})$, the capacitor gets charged and when $v_c(\text{final}) < v_c(\text{initial})$, the capacitor gets discharged.

Based on this criteria, plug-in the values you got in (e) appropriately in the equation for $v_c(t)$ and write down the expression of $v_c(t)$ for-

Charging: $v_c(t) = V_c(9) + [V_c(6) - V_c(9)]e^{-t/\tau}$

Discharging: $v_c(t) = V_c(6) + [V_c(9) - V_c(6)]e^{-t/\tau}$

(g) Now, determine the equivalent resistance as seen from the capacitor terminals. Recall that, while determining equivalent resistance, we must turn off the independent sources.

$R_{eq} = 2 \text{ (k}\Omega\text{)}$

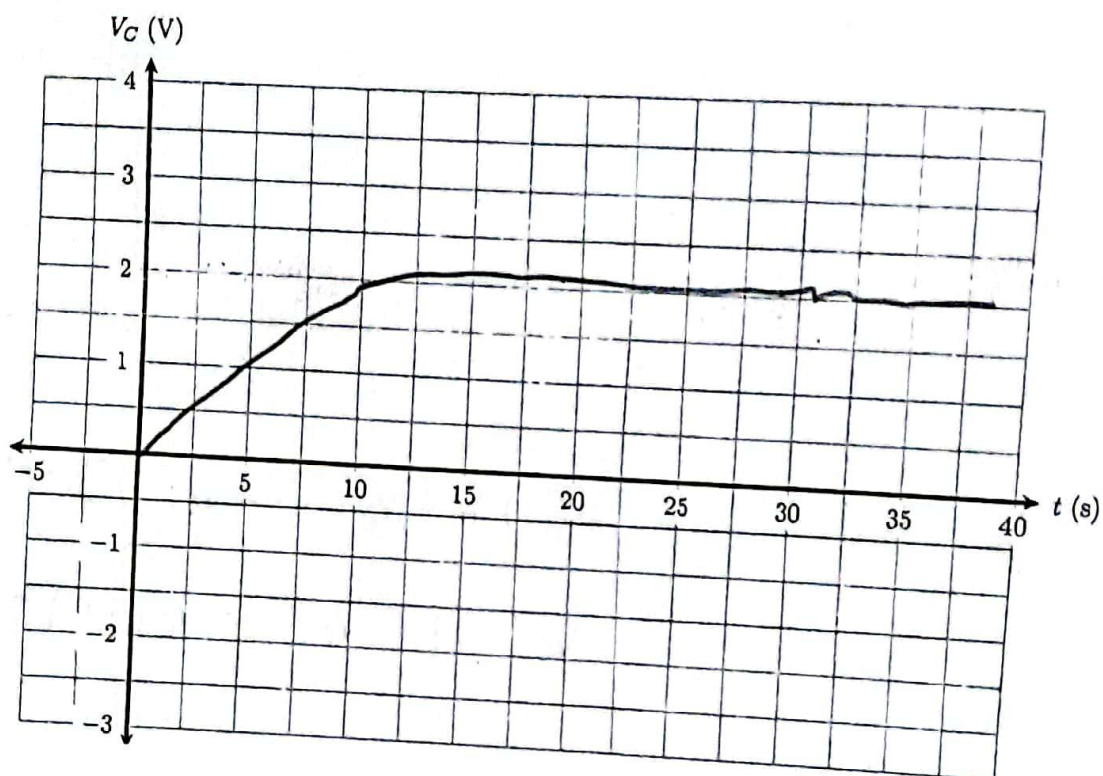
(h) The time constant τ is thus-

$\tau = R_{eq}C = 2$

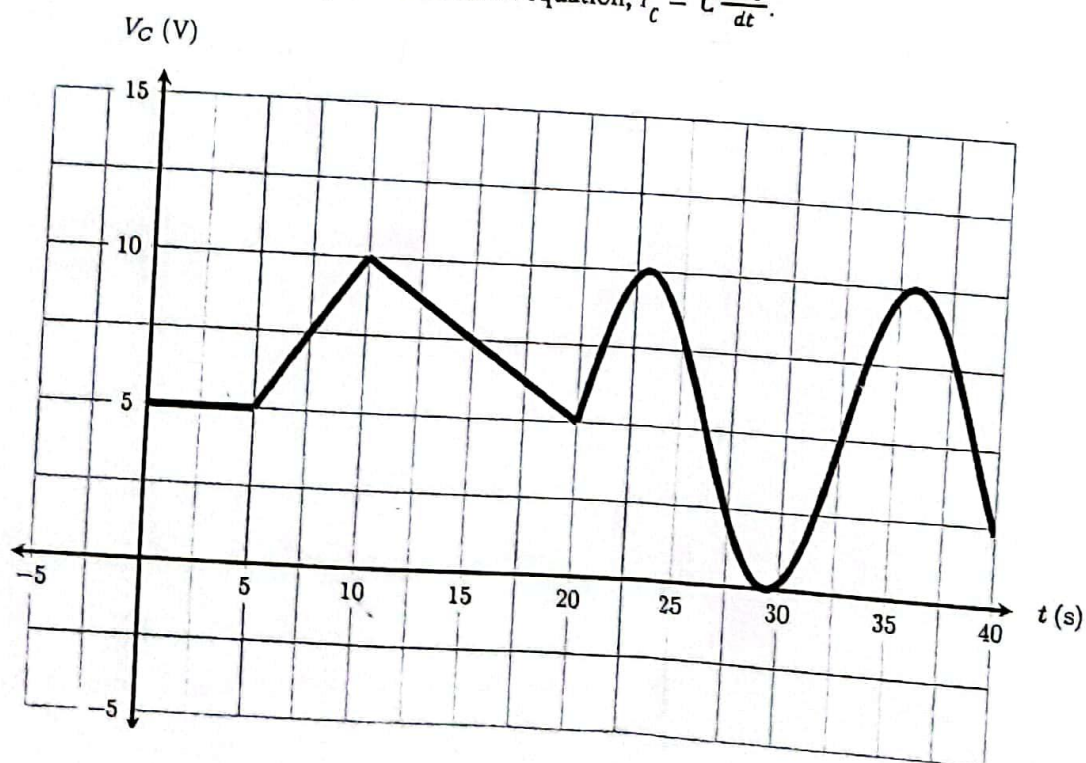
(ms)

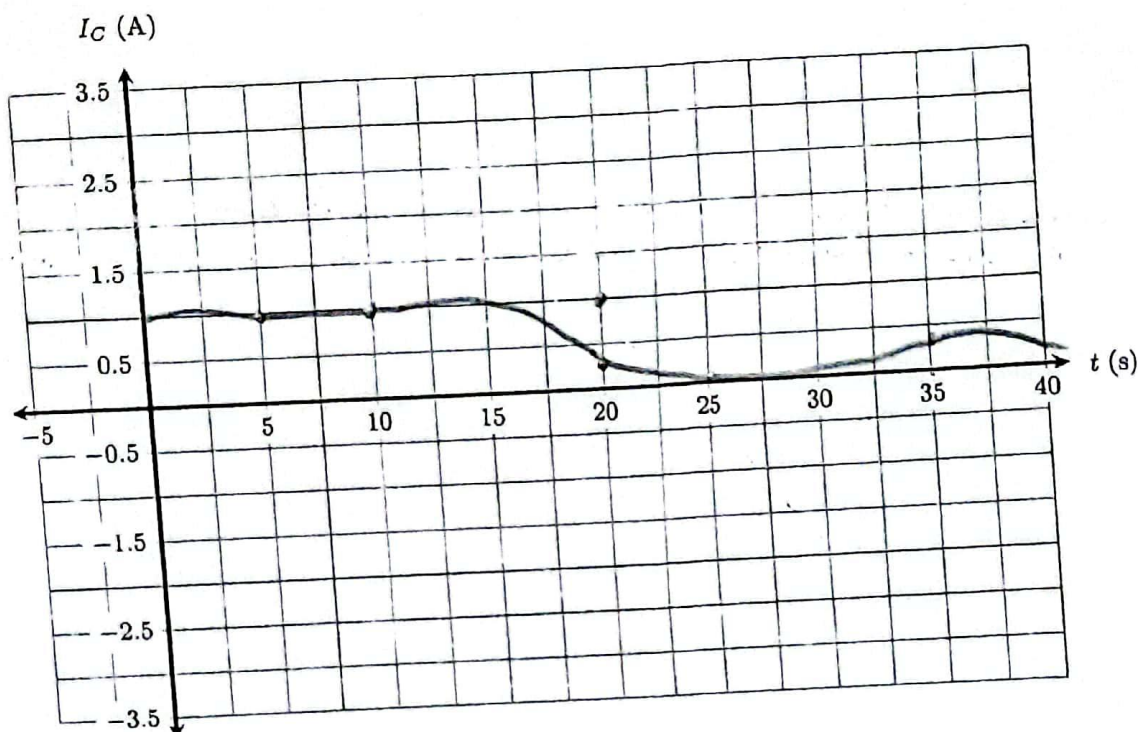
(i) So, it will take 2 (ms) for the capacitor to be fully charged or discharged.

(j) Based on the values in (e), (h), and the frequency (500 Hz), draw the waveform of the voltage across the capacitor v_c that we could observe in an Oscilloscope as a function of time for $t > 0$ as it gets charged and discharged continuously.



9. Draw (in the blank grid provided below) the current through a 1 F capacitor whose voltage is as shown in the following figure. Recall that the voltage (V_C) across a capacitor and the current (I_C) through the capacitor (entering to the '+' polarity of the voltage) are related by the differential equation, $I_C = C \frac{dV_C}{dt}$.





Report

1. Fill up the theoretical parts of all the data tables.
2. Answer to the questions.
3. Attach the captured images of the plots observed in oscilloscope for **Circuits 2, 3** and 4.
4. Discussion [comment on the obtained results and discrepancies]. Start writing from below the line.

There were some task's where we faced difficulties. The obtained results was accurate and after understanding the oscilloscope the rest of the things was much easier.